Global Dynamics Approach to Generative Music Experiments with One Dimensional Cellular Automata

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Abstract

One Dimensional Cellular Automata (CA) offer the use of emergent computation and behaviours as compositional aids to the generative music process. Global dynamics and rule clustering are important concepts in CA research, and CA music research to date has not addressed these topics. Global dynamics and rule clusters offer a new perspective of CA based on the topology of attractor basins. A methodological approach is described as a foundation for future experimentation. Reflective practice techniques will be used as experimental method. Evaluation of results will be judged against recognised criteria. A one dimensional CA is chosen for a foundation experiment in generative music production and visualisations of this will be presented. This work will give CA based generative music a significant shift of context and awareness for the generative artist.

1 Introduction

Algorithmic and computational processes are an important tool for the technology based creative artist producing generative music [3]. CA offer the use of emergent computation and behaviours as compositional aids to the generative art process. CA have been utilised in a number of novel applications in MIDI based computer music [2]. Complexity theory demonstrates that complex systems of simple units, such as the cells in a CA, produce a variety of behaviours. Complex systems such as CA produce global behaviour based on the interactions of these simple units.

CA were conceived by Stanislaw Ulam and John von Neumann in an effort to study the process of reproduction and growths of form [1]. CA are dynamic systems in which time and space are discrete. They may have a number of dimensions, single linear arrays or two dimensional arrays of cells being the most common forms. The CA algorithm is a parallel process operating on this array of cells. Each cell can have one of a number of possible states, sometimes expressed as **k**. The simultaneous change of state of each cell is specified by a local transition rule. The local transition rule is applied to a specified neighbourhood around each cell, sometimes expressed as **r**. The number of cells is given as a number **L**.

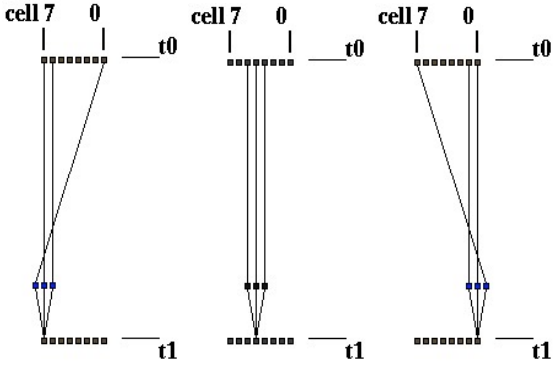


Figure 1. Wiring of periodic boundary cells (left and right) and cell 4 (centre) of an 8 cell (L=8) 1D CA.

CA are usually, but not always, infinite in length. Cells are commonly wrapped around at the edge of the array during the local transition rule computation, to achieve a conceptual infinite array. In this case the array is finite, but unbounded and is said to have periodic boundary conditions. One dimensional binary CA (**k=2, r=1**) have been classed by Wolfram with one of four behaviours [10] as shown in Table 1.

Class 1	**Class 2**	**0lass 3**	**Class 4**
Patterns disappear or become fixed	Patterns evolve to periodic structures	Patterns become chaotic	Patterns grow into complex forms.

Table 1. Wolfram's CA behaviour classes.

The wiring for the edge cells and cell 4 of an 8 cell 1D CA is shown in Fig. 1. Here we can see two time steps of the system from **t0** to **t1**. The transition rule is specified by an 8 bit binary number between 0 and 255 and an example for rule 110 is shown in Fig. 2. The 8 entries in a rule transition table are defined as **T7** to **T0** left to right.

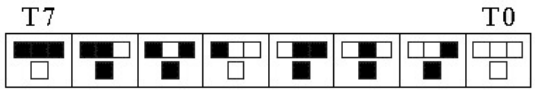


Figure 2. Transition rule table (rule 110).

The λ parameter, introduced by Chris Langton (1986), is an order (Class 1 and 2) – complexity (Class 4) – chaos (Class 3) parameter [7]. This is a kind of virtual tuning knob through the classes of CA rule space. Although the λ parameter appears quite useful, it should be used with care. Langton points

out that it does have weaknesses and will not always be able to work correctly. The λ parameter is seen as a useful tool when dealing with CA of larger **k** and **r**, and future work may benefit from this.

Langton also supported and promoted work on the global dynamics of CA, formally defined by Wuensche and Lesser (1992), which offers a new perspective based on the topology of attractor basins, rule symmetry categories and rule clustering [11]. In this work an atlas of these basins is presented for a variety of small CA sizes up to about 15 cells depending on the particular rule. Here one can compare basin topologies and measures between rules to gain insight into different rule behaviours. Wuensche's Discrete Dynamics Lab (DDLab) software allows for the exploration of global dynamics [13]. A brief overview of Wuensche and Lesser's work on global dynamics, the Z parameter, and rule clustering is presented in the next section.

2 Global Dynamics and Rule Clustering

A CA state space consists of all possible global states. In a finite deterministic CA all state transitions must eventually repeat with period 1 or more. States are either part of an attractor cycle or lie on a transient leading to the attractor cycle. If a transient exists there will be states unreachable by any other states at the extremity. These extremities are called garden of Eden (**goE**) states. All transients leading to an attractor, and the attractor cycle, is termed the basin of attraction (**boa**) of that individual attractor. An example basin of attraction is shown in Fig 3. State space for a particular CA rule and size is populated by one or more basins of attraction, termed the basin of attraction field. The **boa** field may contain **equivalent** basins, where the states of other basins are rotationally symmetric.

DDLab constructs **boa** fields by the computation of pre-images of all states by a reverse algorithm, and can suppress equivalent basins during display. Measures can be taken of the number of basins in the field and garden of Eden states, the attractor periods, basin sizes, maximum transient and cycle period.

Order		Complexity	Chaos
Class 1 : Z = 0.25	Class 2 : Z = 0. 5	Class 4 : Z = 0.75	Class 3 : Z = 1

Table 2. Z parameter for Wolfram's classes.

The Z parameter is an order – complexity - chaos parameter varying from 0 (order) to 1 (chaos) and is calculated from the rule table. There are exceptions to the Z parameter, such as some chaotic behaviour at low Z values. The Z parameter is given for Wolfram's classification in Table 2.

Attractor basin topology reflects the dynamics of a CA rule and can be used as a method of identifying ordered, complex and chaotic behaviour. The **goE** density and in-degree frequency distribution of pre-images are two measures of CA dynamics [12]. The in-degree is the number of pre-images of a node in the **boa** field and is plotted as a histogram. Ordered rules are classified by very high **goE** density and large numbers of high in-degrees. Complex rules will have medium values for both. Chaotic rules have a low **goE** density and large numbers of low in-degrees.

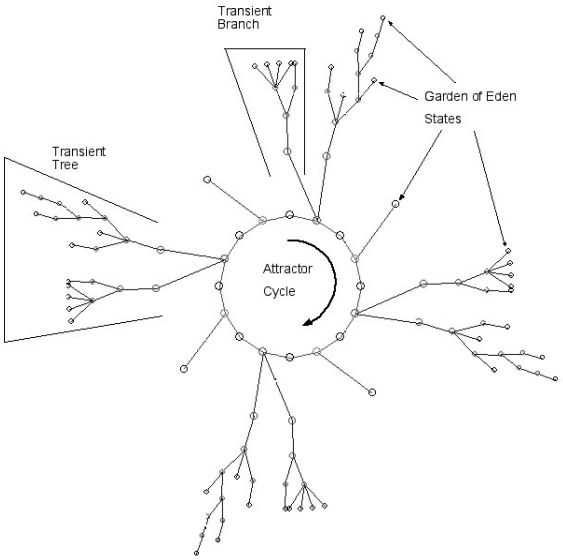


Figure 3. Basin of attraction.

The 1D **k2r1** rules can be grouped into 88 equivalence classes, by **negative**, **reflective** and composite **negative reflective** transforms, with a maximum of four rules per equivalence class [11]. In a rule cluster three rule table transformations are defined as shown in Table 3. The **negative** transform is inversion followed by four pairs of entries having their entries swapped. For example in the first pair **T0** becomes **T7** and **T0** becomes **T7**. The remaining pairs indicated are swapped in the same manner. The **reflection** transform involves pair swapping {**T6** <-> **T3**} and {**T4** <-> **T1**}. The **complementary** transform inverts the rule table contents. The rule cluster axis is shown in Fig 4. and the rule cluster layout is shown in Fig 5.

Negative	Rule table is inverted & T pairs swapped : {**T0** <-> **T7**}, {**T1** <-> **T6**}, {**T2** <-> **T5**}, {**T3** <-> **T4**}
Reflection	T Pairs swapped : {**T6** <-> **T3**} and {**T4** <-> **T1**}
Complementary	Rule table entries are inverted

Table 3. Rule Transformations.

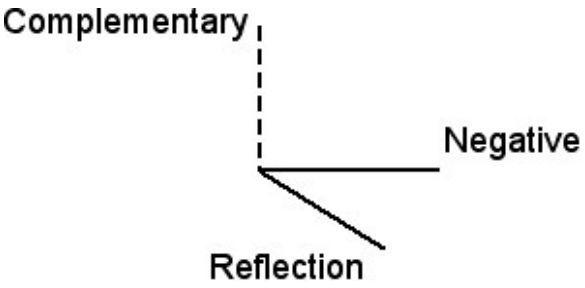


Figure 4. Rule Cluster Axis.

The lowest rule number (**R**) identifies a cluster and is always positioned in the top left corner. The negative (**Rn**) and reflection (**Rr**) transforms are identified, along with a further composite transform the negative reflection (**Rnr**). The complement transform (**Rc**) also has corresponding negative (**Rcn**), reflection (**Rcr**) and negative reflection (**Rcnr**) transforms.

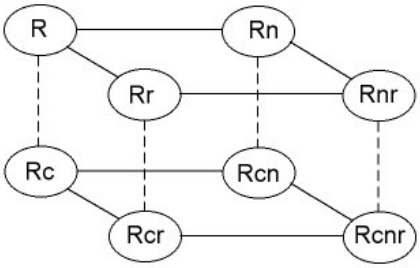


Figure 5. Rule cluster layout.

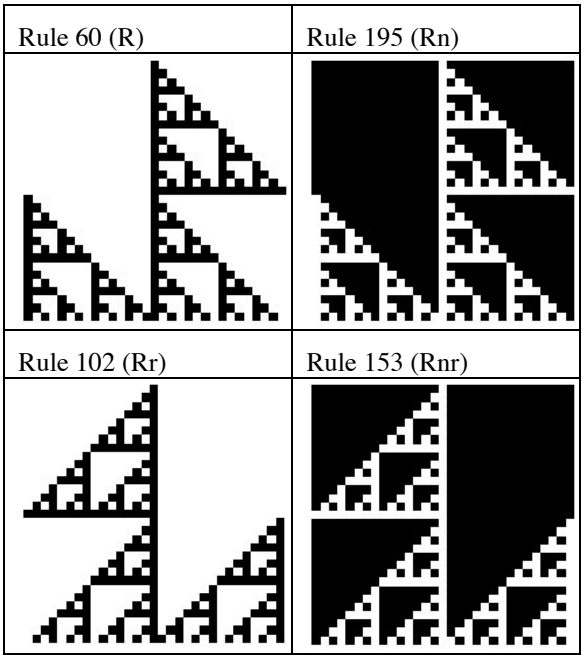


Figure 6. Rule 60 equivalent spacetime patterns.

Each rule cluster has two **boa** fields, one for the top layer (**R**) and one for the complementary bottom layer (**Rc**). Rule clusters contain 2, 4 or 8 different rules depending on whether the transformations result in the same rule number. For both layers the other transformed rules have identical **boa** measures and the states are simply related by being negative (**Rn**), mirror image (**Rr**) or both (**Rnr**), as shown in Fig 6. In some cases the clusters collapse further leaving no complementary rules, resulting in a single **boa** field. The symmetry categories are symmetrical, semi-asymmetrical and fully asymmetrical contained in a total of 48 rule clusters. Symmetric rules contain 2 or 4 rules per cluster, Semi-asymmetric rules have no collapsed clusters and always contain 8 rules. Fully asymmetric rules contain 4 or 8 rules per cluster. Examples of one of each possible cluster type for all categories is shown in Figs. 7a-7c.

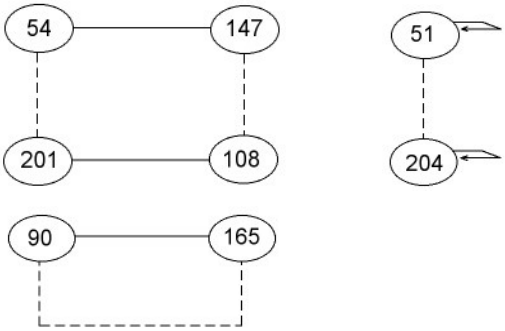


Figure 7a. Symmetric rule clusters, three types.

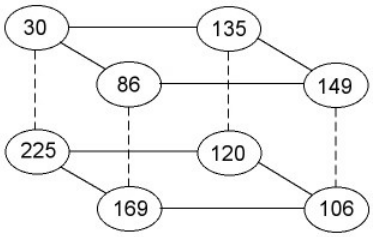
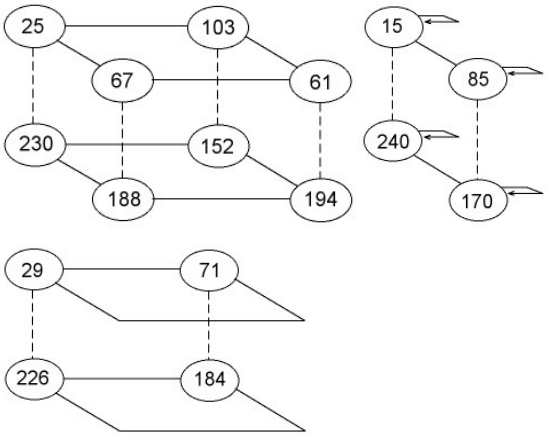


Figure 7b. Semi-asymmetric rule clusters all have this type.



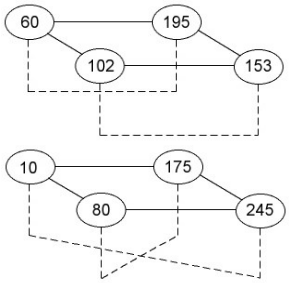


Figure 7c. Fully asymmetric rule clusters, five types.

Rule cluster 15 is important because it exhibits behaviour that is directly comparable to that of analogue sequencers. The cluster diagram for rule 15 is shown in Fig 7c. Rules 170 and 240 are known as the left and right shift rules, and form the complement part of the cluster. Any input pattern will be shifted left (rule 170) or right (rule 240) by one cell per CA step. Analogue sequencer modules commonly have 8, 12 or 16 steps and can be stepped sequentially one stage at a time from between 1 and the maximum number of steps available. This operation is from left to right and vice versa. This is analogous to rule 170 and 240 with a single active cell.

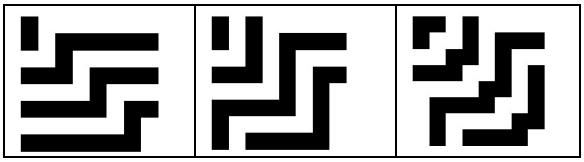


Fig 8a. Rule 15 (wobbly right) L = 8 with seeds of 128, 160 and 208

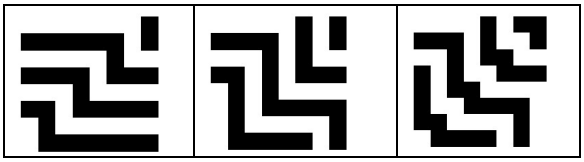


Fig 8b. Rule 85 (wobbly left) L = 8 with seeds of 1, 5 and 11

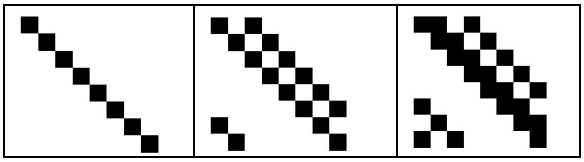


Fig 8c. Rule 240 (right shift) L = 8 with seeds of 128, 160 and 208

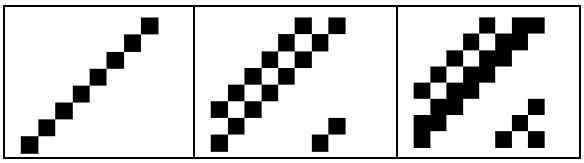


Fig. 8d. Rule 170 (left shift) L = 8 with seeds of 1, 5 and 11

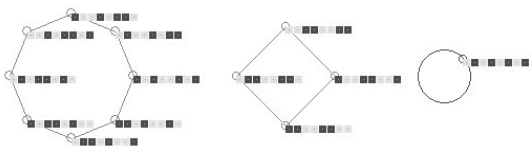


Fig. 9a. Three example basins for rule 15 (wobbly right) L = 8.

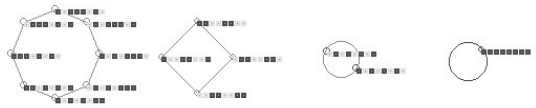


Fig. 9b. Four example basins for rule 240 (right shift) L = 8.

Example spacetime diagrams for each rule are shown in Figs 8a – 8d. In left shift and right shift rules the cycle length is never greater than L. Rules 15 and 85 have this shifting tendency but patterns appear "wobbly" and for some values of L, cycle length can be greater than L.

For rule cluster 15 all states exist on cycles and thus there are no goE nodes, and no transients or preimages on trees. Some example basins are shown in Figs 9a and 9b. Both rules are Z = 1 but appear to be ordered rules and are not chaotic as the Z parameter suggests.

3 Methodology

Reflection-in-action is a method of researching within a practical context [8]. The structure of reflection-in-action in practice based research, termed reflective practice, suggests move testing experiment as a useful methodology. In move testing an action is undertaken to produce intended change. Good results will affirm that move and bad results will negate it. Importantly unintended results that are good, should also affirm the move. Reflection on outcome and potential for future directions will be discussed.

Technical criteria for evaluating sound synthesis techniques were suggested by Jaffe (1995), however these are recognised as guides not rigid definitions [6]. These criteria were suggested in the context of sound synthesis, however some of the criteria applies to generative music production with MIDI. The most appropriate criteria in this context will now be discussed. How efficient is the algorithm? Defined as an extremely important, yet context dependant criterion. Three efficiency categories suggested are memory, processing and control stream. How sparse is the control stream? Related to efficiency and has particular impact on a real time system. How robust is the sounds identity? Asks if parametric adjustment alters the sound too extremely. What classes of sounds can be represented? Conversely this asks if a broad range of sound classes can be achieved by parametric adjustment.

Castagne and Cadoz (2003) revisited and extended these guideline criteria in the context of physical modelling sound synthesis [4]. The context of physical modelling is relevant because it addresses the entire musical creation process and does not just represent a sound synthesis perspective. Ten criteria for evaluating physical modelling were defined in

four areas : computer efficiency, phenomenology, usability of scheme and environment for using the scheme. The efficiency area is similar to Jaffe's proposed scheme. An extended phenomenological criteria is diversity of context and relates to Jaffe's robustness and sound class criteria. In usability particular emphasis is placed on modularity. Modular principles are a very important criterion in obtaining generality, power and simplicity. The environment for using the scheme has two criterion. The first studies if generation algorithms already exist and how effective they are. The second asks if there is a friendly musician oriented environment for using this scheme. Results of this work will be judged on these criteria.

4 Generative Music Experiment

The move testing generative music experiment is defined as : **Action** : Construct visualisations and Max patch with a small, single rule CA. **Intended Change** : Simple, easy to use system with future scalability.

The Max patch was implemented with an 8 cell rule 30 using Bill Vorn's 1D CA Max external. This is part of Vorn's LifeTools, available on the internet, and forms part of the IRCAM library of Max objects [9]. Rule 30 has been chosen simply because this is the default rule in Vorn's external. Once the Max patch is built as a standalone application this will be the only rule available. Changing the rule is possible when using the patch within Max. A CA size of 8 cells was chosen to directly map to a simple 8 voice system.

The approach taken in building an experimental music system has some similarity to the University of York's Cellular Automata Workstation, for the Atari ST [5]. The work described here represents a different approach due to the global dynamics perspective, and the system is viewed as a modular foundation for future work.

A simple realtime system, named CA Simplistic Selector (CASS), was produced to map the CA cells to a set of adjustable MIDI events. This set of events will be termed a Cellular MIDI Event (CME) module. The system thus comprises of 8 CME modules, each module is connected to a unique corresponding cell. A block diagram of the experimental system is shown in Fig. 10.

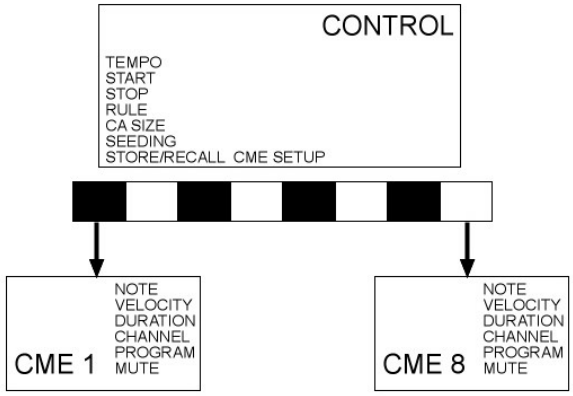


Figure 10. Block diagram of CASS experimental system.

Each CME is controllable in real time by the performer. The CME comprise of note, velocity, duration, MIDI channel and program controls. The system can thus be split up over MIDI channels as desired for multitimbral or one channel for single voice operation. A keyboard for each CME allows for note selection and playing within a seven octave range, a dialogue allows for note input for the full 0 – 127 range. MIDI velocity, channel and program can be input over their complete ranges. Duration is entered by a slider or dialogue from 10 to 10,000 milliseconds.

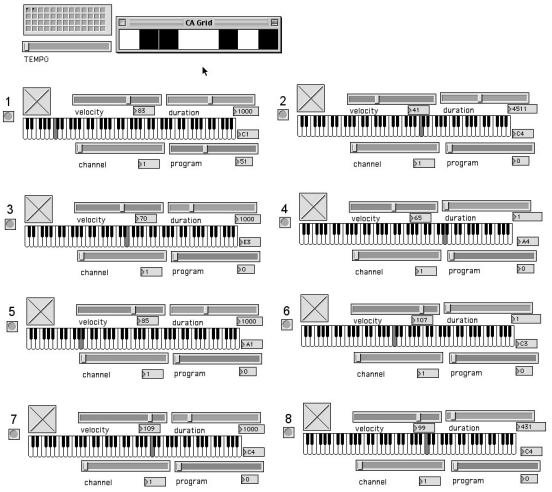


Figure 11. CASS screenshot.

The event onsets are controlled by the 1D CA, which may be seeded randomly, with a list or by a mouse. Individual CME's can be muted during performance. The performer has overall control of tempo and modifies the events to be triggered while the CA runs. This presents an interesting juxtaposition between performer and automated machine. A screen shot of the CASS system is shown in Fig. 11 and a close up of the CME module is shown in Fig 12. All of the CME settings can be saved and recalled for the whole system with the Max preset object.

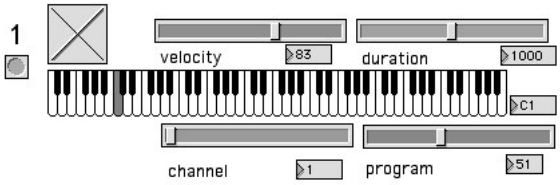


Figure 12. CME module.

The I/O functionality of Vorn's CA Max external is shown in Fig. 13. The inputs have been categorised as seed, clock and data. A number of seeding options are available and the output can be taken in the form of bangs from each cell or dumped as a list of active cells to outlet 2. Resetting of the CA by message is possible, either to an initial seed or to 0 for all cells.

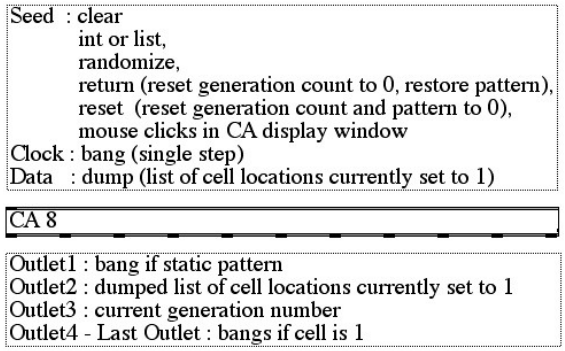


Figure 13. I/O functionality of Vorn's CA Max external.

The CME modules are wired directly to the cell outputs. The CA external will send out a bang from outlet 1 if the output remains static. This allows for the inclusion of automatic reseeding mechanisms to be built within a patch. The current generation count is output as an integer from outlet 3.

Rule 30 is from the semi-asymmetric category and the cluster is shown in Fig. 7b, in Section 2. A boa field for the eight cell Rule 30 was constructed with DDLab and contains 256 states. The main boa is of period 40 and contains 224 states as shown in Fig 14a. The longest transients in this boa contain 16 states and 2 states for the shortest. The remaining states are contained within four boa and are shown in Fig. 14b labelled in decimal. These are a period 8 attractor, two single state period 1 attractors and a two state period 1 attractor. DDLab gave the Z parameter as 1, indicating a chaotic rule. The boa field data was saved to a file and some of the measures obtained are shown in Table 4. The measures in Table 4. agree with the values in Wuensche and Lesser's atlas [11].

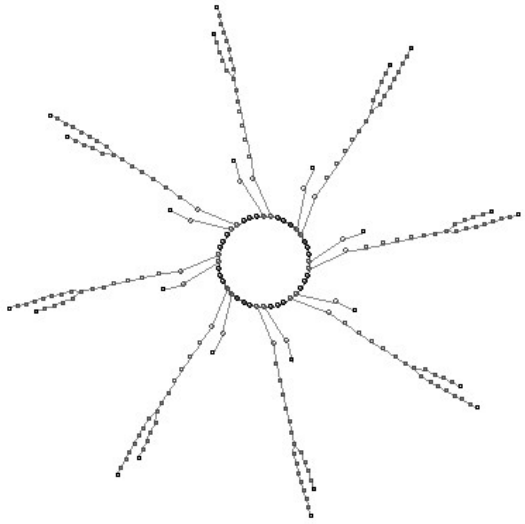


Figure 14a. Eight cell rule 30 basin of attraction period 40.

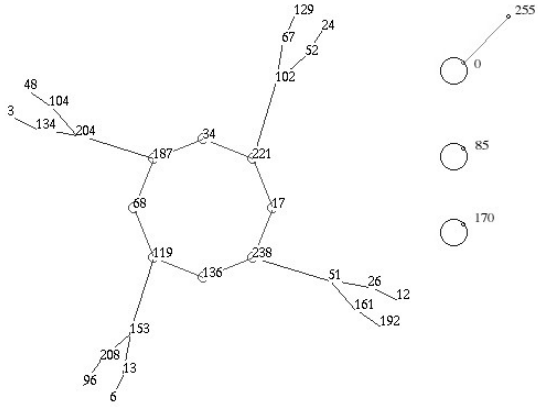


Figure 14b. Eight cell rule 30 basins of attraction with nodes labelled in decimal, period 8 (left) and 3 x period 1 (right).

type	no. basins	period	no. states	% of states	goE	max transient
1	1	1	2	0.781	1	1
2	1	40	224	87.5	24	16
3	1	8	28	10.9	8	3
4	2	1	1	0.781	0	0

Table 4. Summary of eight cell rule 30 boa measures.

DDLab can plot the state space matrix in decimal, showing the state space in terms of the left and right half of each states bit string. This results in an 8x8 grid showing every possible state in the field. The state space matrix of the entire field of rule 30 is shown in Fig. 15a. with each basin assigned a different shading. The state space matrix for attractor cycles only is shown in Fig 15b, with the same shading scheme.

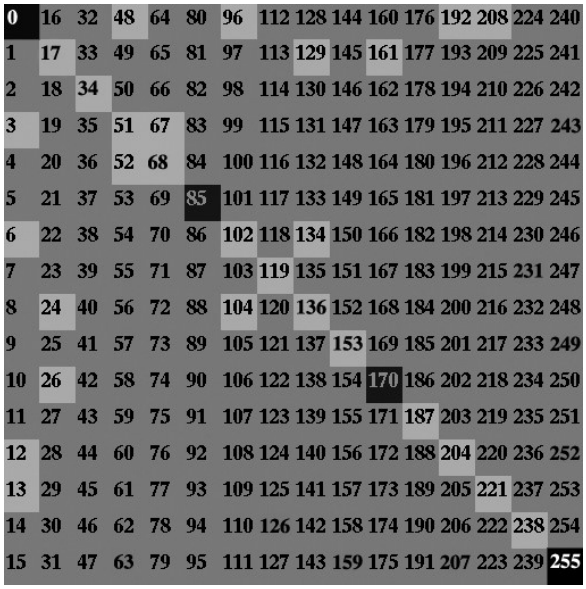


Figure 15a. Eight cell rule 30 state space matrix, all states.

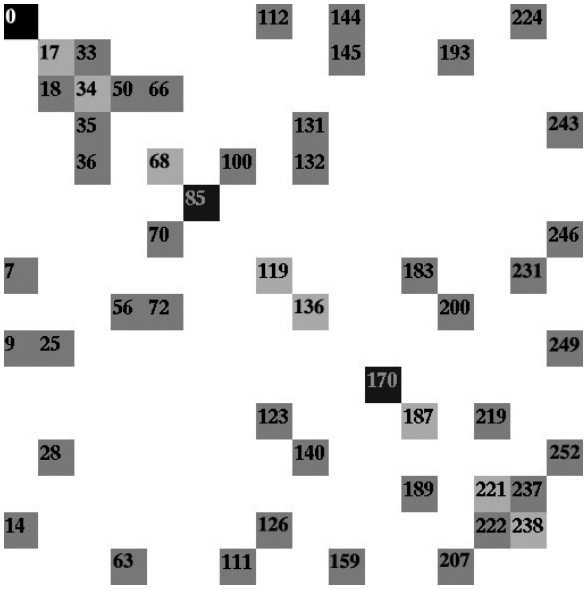


Figure 15b. Eight cell rule 30 state space matrix, attractor cycle states only.

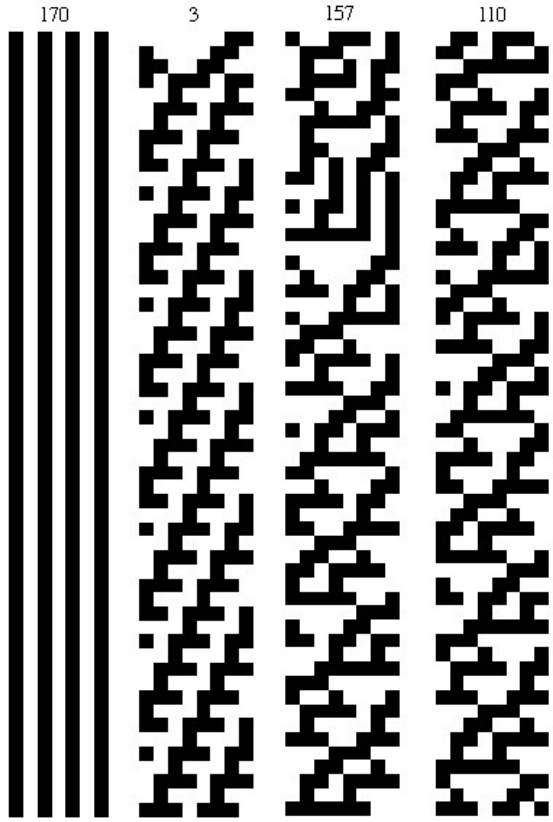


Figure 16. Spacetime plots of 8 cell Rule 30 with decimal seed states indicated above.

Spacetime plots of three attractor cycles from four example seed states are shown in Fig 16. Space is the horizontal axis and time progresses vertically downwards for 56 generations. Running the system for 56 generations allows for the maximum transient and cycle period to appear. From left to right these plots are; Period 1 (type 4) with seed = 170, period 8 (type 3) with goE seed = 3, longest transient to period 40 (type 2) with goE seed = 157 and shortest transient to period 40 (type 2) with goE seed = 110. The period 40 patterns show how the cycle will be entered at a different phase, and is also seen topologically on the boa diagrams. Generally speaking the period 8 and period 40 patterns appear to move right to left over time. The period 8 attractor looks reasonably ordered, whereas the period 40 visually has a slightly more random appearance in comparison.

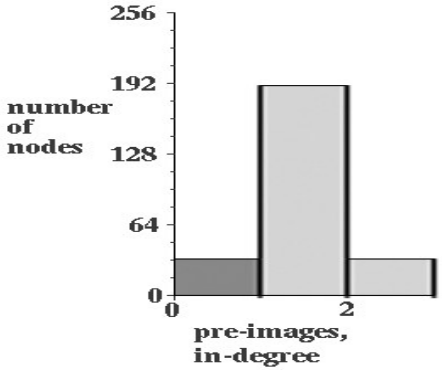


Figure 17. In-degree histogram for 8 cells rule 30, indicating a chaotic dynamic.

A pre-image in-degree histogram is shown in Fig. 17. The goE nodes with zero in-degree pre-images are in the left column, the remainder of the field has 1 or 2 in-degree pre-images. There is a low goE density, and high frequency of low in-degrees indicating by Wuensche's (1997) classification that this is a chaotic rule[12].

5 Evaluation and Reflection

Experimental results are now evaluated in terms of the four criteria areas previously introduced in section 3; computer efficiency, phenomenology, usability of scheme and environment for using the scheme. Reflections will also be given on the system in use as part of the generative composition process.

In terms of memory the 8 cell CA is very efficient. The CASS system is able to run and respond to user events in realtime over a broad tempo range. Vorn's CA external will not run in Max overdrive mode and this results in interruptions in MIDI output scheduling during some GUI events. This does not adversely affect studio use, but is sometimes noticeable in a performance context. Overall the control stream is light and messages are not sent when a cell is not active. This MIDI control stream is minimised by the simple and limited nature of each CME module. MIDI Channel and Program Change are only sent when changed by the user. MIDI Continuous Controllers such as Pitchbend, Mod Wheel and Pan have subsequently been informally tested without much additional efficiency overhead.

In terms of phenomenological criteria, successful diversity of application context was achieved, as was robustness. The main quality of the system is a temporal mapping of the CA to specified sounds. In the built application the seed state and tempo are the performance parameters affecting the CA. From random seed conditions the system is most likely to enter the period 40 basin. In general though the system will tend to be in the period 40 or period 8 basins from a random start, and the period one basins are not often reached randomly.

The aesthetic results produced during the composition process up to date have been both rhythmic and textural. Two compositions emerged from these experiments, exemplifying these approaches. "Ubendem, Wemendem" which uses rhythmic elements in the form of percussive sounds triggered by CASS. The majority of rhythm in this piece was created by selecting sounds while the system ran and storing them to build up a few simple patterns. "Cassorgize" is a slower tempo textural piece, using filtered organ sounds with a slow attack and long release. This approach involved selecting sounds first by assigning event parameters and then running the CASS system from a variety of different seed states.

Modular principles are a very important usability criterion in obtaining generality, power and simplicity. CASS has been successful by taking a basic modular approach. The modularity of this system could easily be extended by the addition of a simple switching matrix between the CA cells and the CME modules. This would allow for a single cell to trigger multiple CME's. The system should be able to scale easily both in CA size and the number of CME's, but this will have a bearing on efficiency criteria. It should be noted that when investigating the global dynamics with DDLab for larger numbers of cells there is a practical limit. The suggested limit in the manual is 16 cells for boa fields and 18 cells for a single boa [13].

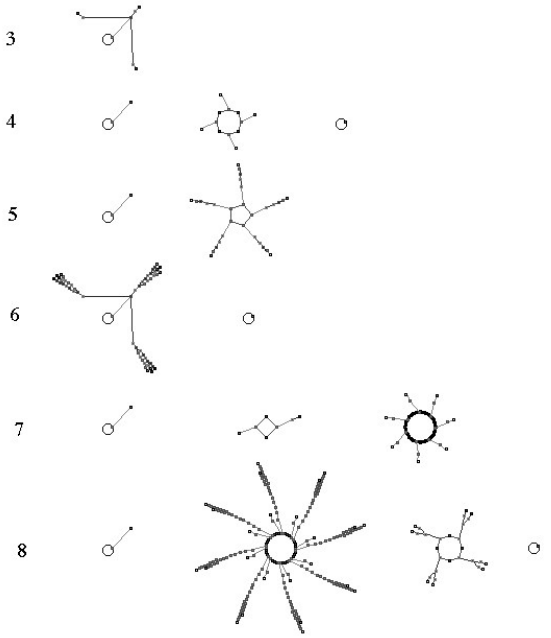


Figure 18. Rule 30 boa fields for 3 to 8 cells, equivalent basins are suppressed.

Rule 30 boa fields for sizes of 3 to 8 cells are shown in Fig. 18. A variety of basin types can be seen and DDLab allows for equivalent rotationally symmetric basins to be suppressed from the display. It will be interesting to examine both increasing and decreasing the number of cells compared to the current system. Examples of the boa fields for 9, 10 and 11 cell rule 30 are shown in Fig. 19.

Some I/O problems have been identified in the context of modular based generative music systems. The seed messages are passed as a list of active cell numbers and having an integer or binary seed is preferable. The cell output of the system would also benefit from being available in this format. Vorn's CA is currently adjustable from 4 to 256 cells, which is perfectly adequate. However, changing the number of cells involves changing the argument of the external within the Max patch. Similarly the rule can only be changed when the patch is being edited, using the Get Info command. The system would

benefit by having these parameters available as inputs.

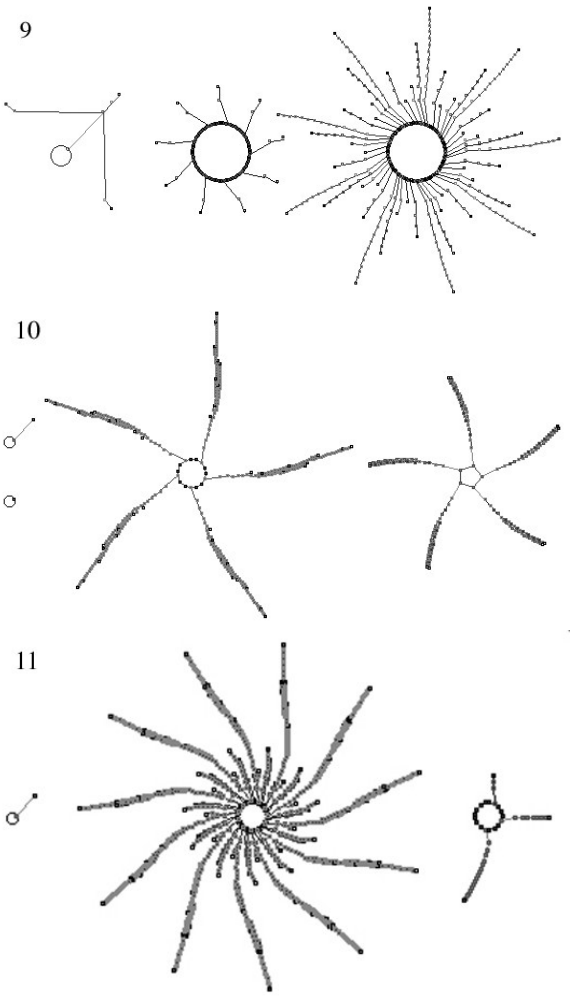


Figure 19. Rule 30 boa fields for 9, 10 and 11 cells, equivalent basins are suppressed.

The overall environment for using the scheme is simple and easy to use. Generation algorithms exist in the context of both visualising CA dynamics, and within the Max programming environment. Both have been effective for generative music research and production with the CASS 8 cell experimental system. Global dynamics is easily studied on a variety of computer platforms with DDLab for systems with small numbers of cells. DDLab has the added benefit of producing basin measures and diagrams in one software environment. Preparation of Fig. 8 and 16 spacetime diagrams to confidence check Vorn's CA external was aided by the ability to label basin nodes with state values, both in decimal and binary.

State space matrix plots are useful and offer a clear numerical view of the basin field. Spacetime diagrams can be easily produced in both DDLab and Mathematica 5. The Max programming environment allows for the investigation of experimental systems within a live and studio context. Vorn's CA is a Mac OS9 based external, currently untested and unlikely to work for OSX. This does not present a problem at present but future work will require an OS9 machine to be available or a new OSX based module. A 1D CA is available in Jitter, but this does not appear to implement periodic boundary conditions and an upgrade from Max is also required.

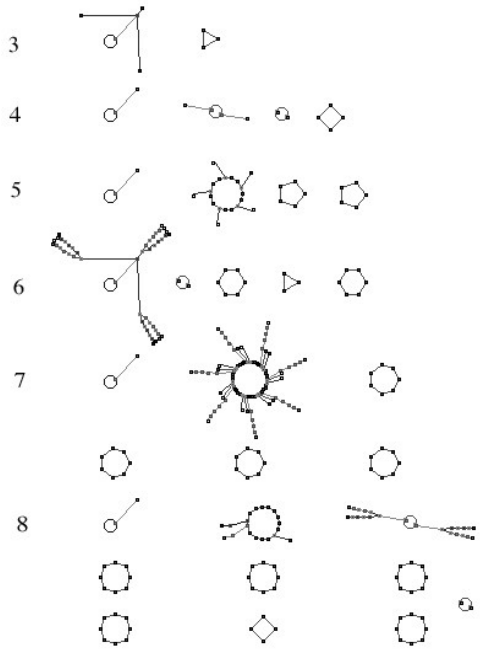


Figure 20. Rule 225 boa fields for 3 to 8 cells, equivalent basins are suppressed.

6 Conclusion and Future Work

In using CA for generative music we show that successful and sensitive application requires understanding of the science by the artist. Global dynamics and rule clustering are important concepts for the generative artist intent on using CA. The vast behaviour space of 1D CA will remain an untamed wilderness for music application unless these concepts are acknowledged, and actively researched in a musical context. The methodology presented in this paper will serve as the foundation for future music research in modular CA systems. Art and science are different yet fundamentally connected disciplines. Here the science has inspired and assisted the artist to achieve specific results. It is hoped that the artwork produced will inspire the science community and generate better understanding in that community of the musical applications of CA.

Vorn's CA external was created for his robotic art systems and not as a modular music component. The criticisms of its functionality are to be taken based on criteria for music composition, and not in terms of his individual application. The current work is made possible because of his placement of them within the public domain.

Rule cluster 15 is a fundamental connection to the traditional analogue sequencer and provides a

familiar departure point for electronic musicians. Further work is planned at CCS on generative music research with other 1D CA and will be presented as it is completed. This will continue in a modular context with other parametric mappings and a selection of rules, including rule 30 and its complement rule 225 shown in Fig. 20. The CASS system has been used at Noyzelab and CCS in both performance, and studio production of generative compositions. It is a simple to use musical interface to CA dynamics, both within the Max environment and as a standalone application.

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